



Assessment Analysis Guide
25-26 Math G8 Form D Sci Notation
Mathematics
Grade 8

Reference #: 683613a

1

Item Id: i184999

Item Type: SR (Selected Response)

Standards Description:

8.EE.A.1 Know and apply the properties of integer exponents to generate equivalent numerical expressions.

Content:

Which expression is equivalent to the one below?

$$(2^6 \times 9^6) \div 18^{14}$$

A $\frac{1}{18^{-2}}$

B $\frac{1}{18^8}$

C 18^8

D 18^{20}

Primary Standard: 8.EE.A.1**Points Possible:** 1**Distractor Rationale:**

A: $\frac{1}{18^{-2}}$

Student multiplied the bases and added the exponents when simplifying the dividend instead of multiplying the bases and keeping the common exponent, then subtracted the exponent in the divisor from the exponent in the dividend and misunderstood how negative exponents relate to division.

B: $\frac{1}{18^8}$

Correct.

C: 18^8

Student correctly simplified the dividend by multiplying the bases and keeping the common exponent, but subtracted the exponent in the divisor from the exponent in the dividend in the wrong order, OR successfully simplified to a negative exponent but did not recognize the need to use the reciprocal to make it positive.

D: 18^{20}

Student correctly simplified the dividend by multiplying the bases and keeping the common exponent, but added the exponents in the divisor and the dividend instead of subtracting.

2

Item Id: i185005

Item Type: SR (Selected Response)

Standards Description:

8.EE.A.4 Perform operations with numbers expressed in scientific notation, including problems where both decimal and scientific notation are used. Use scientific notation and choose units of appropriate size for measurements of very large or very small quantities (e.g., use millimeters per year for seafloor spreading). Interpret scientific notation that has been generated by technology.

Content:

Light travels at a speed of 1.9×10^5 miles per second.

How far, in miles, does light travel in 30 seconds?

A 5.7×10^1

B 5.7×10^4

C 5.7×10^6

D 5.7×10^7

Primary Standard: 8.EE.A.4**Points Possible:** 1**Distractor Rationale:**

A: 5.7×10^1

Student multiplied the coefficient and the multiplier but did not apply the exponent when performing the calculation.

B: 5.7×10^4

Student mistakenly used a negative exponent instead of a positive exponent when working with powers of ten, leading to an incorrect place value adjustment.

C: 5.7×10^6

Correct.

D: 5.7×10^7

Student thought the exponent meant how many zeros to add when converting to standard form.

3

Item Id: i184994

Item Type: SR (Selected Response)

Standards Description:

8.EE.A.1 Know and apply the properties of integer exponents to generate equivalent numerical expressions.

Content:

Which expression is equivalent to the one below?

$$\frac{4}{(4^2)^5}$$

A 4^{-10}

B 4^{-9}

C 4^{-7}

D 4^{11}

Primary Standard: 8.EE.A.1**Points Possible:** 1**Distractor Rationale:**

A: 4^{-10}

Student correctly simplified the power to a power but thought the exponent in the numerator was 0 and subtracted instead of recognizing the existing exponent.

B: 4^{-9}

Correct.

C: 4^{-7}

Student added the exponents in the denominator instead of multiplying, did not account for the base in the numerator, and rewrote the fraction that had an exponent in the denominator using a negative exponent.

D: 4^{11}

Student correctly simplified the power to a power but added the exponents in the numerator and denominator instead of subtracting.

4

Item Id: i185006

Item Type: SR (Selected Response)

Standards Description:

8.EE.A.4

Perform operations with numbers expressed in scientific notation, including problems where both decimal and scientific notation are used. Use scientific notation and choose units of appropriate size for measurements of very large or very small quantities (e.g., use millimeters per year for seafloor spreading). Interpret scientific notation that has been generated by technology.

Content:
Simplify.

$$\frac{6.0 \times 10^8}{1.2 \times 10^5}$$

- A

4.8×10^3
- B

5.0×10^3
- C

5.0×10^{13}
- D

7.2×10^{13}

Primary Standard: 8.EE.A.4

Points Possible: 1

Distractor Rationale:

- A:

4.8×10^3

Student correctly subtracted the exponents but subtracted the coefficients instead of dividing them.
- B:

5.0×10^3

Correct.
- C:

5.0×10^{13}

Student correctly divided the coefficients but added the exponents instead of subtracting them when dividing powers of ten.
- D:

7.2×10^{13}

Student added the coefficients instead of dividing them and also added the exponents instead of subtracting them.